Assignment 1: Data Profiling (EDA) Report (10 marks)

Objective

Show how you profile a dataset and deal with the data problems you find. This assignment is about **diagnosing and fixing data quality issues**, not about presenting insights or serving data to an audience.

Task

Find any real-world scenario that fits the situations below, meaning the data has missing values, outliers, quality issues, and possible PII or bias risks. Take data for that scenario from a public API, a scraped web page, or an open dataset (e.g., Kaggle, data.gov.my). You can also use your capstone project data. Work in a Jupyter notebook and submit the link to your GitHub repository. Please include all your team members and their matric number.

Your notebook must include

1. **Scenario:** the scenario you found, why it fits the situations above, and the questions you want to answer (2 to 3 sentences)
2. **Data source:** where the data came from, with the URL and how you obtained it
3. **Data profiling and issue handling:**
    - **Missing values:** where and how much, and how you handled them (drop, impute, flag), with your reason
    - **Outliers:** how you detected them (e.g., IQR, z-score, boxplot) and whether you kept, capped, or removed them, with your reason
    - **Other quality issues:** duplicates, wrong data types, inconsistent formats, invalid values, and how you fixed them
4. **Exploratory (diagnostic) charts:** at least 2 labeled charts that expose data problems, such as a missing-value heatmap, boxplots or histograms for outliers, value counts for inconsistent categories, or group balance.
5. **PII and bias risks:** sensitive or personally identifiable fields, sampling bias, or unbalanced groups, and how you would handle them
6. **Conclusion:** whether the data is fit for the scenario and what limitations remain

GitHub requirements

- Send only the link from your GitHub that contains the Jupyter notebook (.ipynb) which must run from top to bottom-include the scenario, data source, and instructions to run the notebook

Tips

- Pick a scenario whose data is messy enough to show real problems. A perfectly clean dataset gives you little to profile and will earn fewer marks.
- Explain the reasoning behind every decision. For example, "I dropped these rows because..." provide justification than simply dropping them.
- Respect each website's terms of use if you scrape data.